MAT 017B Midterm 2

August 30, 2018

Name:____

SID:____

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1 \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f\left(x_k\right)\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x) \qquad \frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f[g(x)]g'(x) - f[h(x)]h'(x)$$

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C \qquad \int u\,dv = uv - \int v\,du$$

$$A = \int [f(x) - g(x)]\,dx \quad \Delta y = \int \frac{dy}{dx}\,dx \quad \bar{y} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$f(x) \approx \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \quad \frac{dy}{dx} = f(x)g(y) \quad \int \frac{dy}{g(y)} = \int f(x)\,dx$$

$$y' + p(x)y = q(x) \quad I(x) = e^{\int p(x)dx} \quad y = \frac{1}{I(x)}\int I(x)q(x)\,dx$$